3 Unique Skills Per Role

Skills that belong to one role and are rarely needed by the other two.

AI Engineer	ML Engineer	Data Scientist
1 Prompt Engineering	1 Model Training & Fine-tuning	1 Statistical Analysis
2 RAG	2 Feature Engineering	2 Exploratory Data Analysis (EDA)
3 Agent & Tool Design	3 MLOps & Model Serving	3 Data Storytelling

AI Engineer

1 Prompt Engineering

WHAT IT IS	Writing instructions (called prompts) that reliably get the LLM to produce the output you actually want. Small wording changes can completely change the result, so this is a craft in itself.
WHY UNIQUE	ML Engineers and Data Scientists never need to do this because they do not interact with LLMs directly in their day-to-day work. This is the AI Engineer's primary tool.
EXAMPLE	*Changing a prompt from 'Summarise this email' to 'Summarise this email in 3 bullet points for a busy manager who needs to decide in 30 seconds whether to reply' — and getting a much more useful result because of it.*

2 RAG (Retrieval-Augmented Generation)

WHAT IT IS	Connecting an LLM to your own data (documents, databases, websites) so it can answer questions about content it was never trained on. The model looks up relevant chunks, then uses them to write its answer.
WHY UNIQUE	This is a purely AI Engineering pattern. ML Engineers train models on data; they do not build retrieval pipelines to feed live context into LLMs at query time. Data Scientists analyse data rather than wire it to models.
EXAMPLE	*A legal firm uploads 10,000 contracts. You build a RAG system so lawyers can ask 'Which contracts expire in Q3?' and get an instant accurate answer — without retraining any model.*

3 Agent & Tool Design

WHAT IT IS	Building systems where an LLM can take actions — not just reply with text. You give the model tools (search the web, run code, call an API, send an email) and it decides which to use to complete a task.
WHY UNIQUE	ML Engineers build models. Data Scientists analyse data. Neither designs the loop that lets an LLM plan and act autonomously across multiple steps. This is almost entirely an AI Engineer concern.
EXAMPLE	*An agent that receives 'Book me a flight to Berlin next Tuesday under $500' — it searches flights, checks your calendar, compares prices, and books without you touching a browser.*

3 Unique Skills Per Role

Continued — ML Engineer

ML Engineer

1 Model Training & Fine-tuning

WHAT IT IS

Writing the code that teaches a neural network to get better at a task by adjusting its internal weights using labelled examples. Fine-tuning adapts an existing model to a specific domain using your own smaller dataset.

WHY UNIQUE

AI Engineers use models that already exist — they never touch training code. Data Scientists build simpler statistical models (regression, decision trees) but do not train deep neural networks. Training is the ML Engineer's core job.

EXAMPLE

Taking GPT-2 and fine-tuning it on 50,000 medical records so it speaks in the precise clinical language your hospital uses — something a generic model cannot do straight out of the box.

2 Feature Engineering

WHAT IT IS

Transforming raw data into the specific numerical inputs that give a machine learning model the best chance of learning the right patterns. Choosing what to feed the model — and how — has a huge impact on accuracy.

WHY UNIQUE

AI Engineers do not train models, so they never engineer features. Data Scientists explore data but are not responsible for turning it into production-ready model inputs at scale. This lives squarely in ML Engineering.

EXAMPLE

For a house price model, instead of feeding 'date built', you create 'age of house in years' and 'is it older than 50 years'. That small transformation makes the model significantly more accurate.

3 MLOps & Model Serving

WHAT IT IS	Getting a trained model out of a notebook and into production — packaging it as an API, monitoring it for errors, retraining it when its accuracy declines (called model drift), and versioning it so you can roll back safely.
WHY UNIQUE	AI Engineers deploy API calls, not trained models. Data Scientists finish at the insight or notebook stage. Only the ML Engineer is responsible for keeping a live model healthy in a production system over months and years.
EXAMPLE	*A fraud detection model trained in January starts flagging fewer real frauds by March because criminals changed tactics. The ML Engineer detects the drift, retrains on fresh data, and deploys the new version with zero downtime.*

3 Unique Skills Per Role

Continued — Data Scientist

Data Scientist

1 Statistical Analysis

WHAT IT IS	Using probability and statistics to understand what data is actually telling you — and whether what you see is real or just random noise. This includes hypothesis testing, confidence intervals, and correlation analysis.
WHY UNIQUE	AI Engineers work with model outputs, not raw statistics. ML Engineers focus on model accuracy metrics, not business-level statistical inference. Interpreting whether a result is statistically significant is the Data Scientist's unique superpower.
EXAMPLE	*After an A/B test where version B got 3% more clicks, the Data Scientist runs a significance test and finds the result is not statistically significant — the difference could be random chance — so the team does not ship version B.*

2 Exploratory Data Analysis (EDA)

WHAT IT IS	Systematically digging through a new dataset to understand its shape before drawing any conclusions — finding missing values, outliers, unexpected patterns, and relationships between variables. Always done before any modelling.
WHY UNIQUE	ML Engineers assume data is already understood and focus on the model. AI Engineers receive clean structured outputs from APIs. Only the Data Scientist starts every project by asking 'What is actually in this data and can I trust it?'
EXAMPLE	*Given a new sales dataset, the Data Scientist discovers that 10% of revenue records are duplicated (a data pipeline bug), and that sales in one region are 100x higher than others due to currency not being normalised. Both would have broken any model built on top.*

3 Data Storytelling

WHAT IT IS	Translating numbers, charts, and model outputs into a clear narrative that non-technical people can understand and act on. This means choosing the right chart, cutting unnecessary detail, and framing findings around a business decision.
WHY UNIQUE	AI Engineers ship products used by end users. ML Engineers write technical documentation for other engineers. Only the Data Scientist regularly presents findings to executives, product managers, and other non-technical stakeholders who need to make decisions based on the analysis.
EXAMPLE	*Instead of showing a table of 200 churn rate numbers, the Data Scientist builds one slide: 'We are losing users aged 25-34 in the south-east region at 3x the normal rate. If we fix the onboarding flow there, we recover an estimated $2M/year.' The VP approves the project immediately.*

All 9 Skills — Summary

Role	#	Skill	One-line description
AI Engineer	1	**Prompt Engineering**	Write instructions that reliably guide the LLM's output
	2	**RAG**	Connect an LLM to your own documents so it can answer questions about them
	3	**Agent & Tool Design**	Let an LLM take actions (search, run code, call APIs) to complete tasks
ML Engineer	1	**Model Training & Fine-tuning**	Write code that teaches a neural network on your dataset
	2	**Feature Engineering**	Transform raw data into numerical inputs that make models accurate
	3	**MLOps & Model Serving**	Deploy a model to production and keep it healthy over time
Data Scientist	1	**Statistical Analysis**	Determine whether a result is real or just random noise
	2	**Exploratory Data Analysis**	Understand a new dataset — find problems before they break the model
	3	**Data Storytelling**	Turn numbers into a clear narrative non-technical people can act on